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School of Computer Applications · Department of BCA (DS & AI)

BUSINESS ANALYTICS · GRADED ASSIGNMENT

Descriptive Analytics Report

Zomato Sales & Restaurant Behaviour in India

State-wise Distribution · Cuisine · Rating · Pricing · Online Delivery

Product / Service analysed: Zomato (online restaurant discovery & food delivery platform)

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1. Executive Summary

This report uses descriptive analytics to show how restaurants on **Zomato** are spread across India and how they behave on the platform. It covers five themes: **state-wise distribution, cuisine, rating, pricing and online delivery**. The data is the public Zomato Restaurants dataset (Kaggle). Of its 9,551 listings, 8,652 are Indian restaurants in 43 cities, mapped to 22 states/UTs. The dataset has no order or revenue figures, so **votes** are used as the measure of customer engagement (a sales proxy) and **average cost for two** as the price measure.

Key findings

- **Geography:** Delhi NCR holds 91.9% of listings; Delhi alone has 63.3%.
- **Cuisine:** North Indian (45.6%), Chinese (31.1%) and Fast Food (22.7%) dominate.
- **Rating:** mean 3.35, median 3.3, SD 0.50; 24.7% of restaurants are unrated.
- **Pricing:** median cost for two Rs.450, mean Rs.624. Prices are right-skewed, and 49.6% of listings are in the budget range.
- **Online delivery:** offered by 28.0% of restaurants, most often in the moderate price range (44.8%). Delivery restaurants get 6× the median votes.
- **Engagement:** the top 10% of restaurants receive 68.9% of all votes.

2. Introduction

Zomato, founded in 2008 in Delhi, is one of India's leading platforms for restaurant discovery, reviews and online food delivery. A restaurant's Zomato listing (its rating, votes, price level, cuisines and delivery option) strongly shapes how customers find and choose it. Zomato was chosen because it is widely used and has a large, public dataset. The report is **descriptive**: it summarises what the data shows and does not predict or prescribe.

Table 1. Objectives of the study

#	Objective	Technique
1	Describe the state-wise and city-wise distribution of restaurants	Aggregation, frequency
2	Identify the most common cuisines and compare their price and rating	Frequency, group medians
3	Summarise customer ratings	Mean, median, mode, SD
4	Describe pricing (cost for two, price range)	Central tendency, spread
5	Measure online-delivery availability and compare restaurants with and without it	Proportions, cross-tabulation
6	Relate price, rating, delivery and votes	Cross-tab, heat map, progression

3. Data Collection & Methodology

Source: secondary data, the *Zomato Restaurants Data* dataset (Mehta, 2018, Kaggle), collected through the Zomato API around 2017–2018. It is real data; no values were simulated. **Tools:** Python (pandas, NumPy, matplotlib) for the analysis and charts, and ReportLab for the PDF.

Table 2. Dataset profile

Item	Value
Total records / columns	9,551 / 21
Indian records used	8,652 (90.6%)
Cities / states / localities	43 / 22 / 784
Time period	Cross-sectional snapshot (c. 2017–18), no date field
Unit of observation	One restaurant listing

Table 3. Variables used

Variable	Type	Use
City → State (derived)	Categorical	State-wise and city-wise distribution
Cuisines	Multi-label text	Cuisine frequency (split on commas)
Average Cost for two	Numeric (Rs.)	Pricing
Price range	Ordinal 1–4	Price tiers (budget to luxury)
Aggregate rating / Rating text	Numeric 0–4.9 / ordinal	Rating analysis (0 = not rated)
Has Online delivery / Table booking	Yes / No	Delivery and booking analysis
Votes	Count	Customer engagement (sales proxy)

Limitations

- Delhi NCR makes up 91.9% of the data; most other cities have only about 20 (mostly popular) restaurants each, so their averages are not representative of the whole city.
- There are no order, revenue or date fields, so votes are used as a proxy for sales and a time trend is replaced by a progression across price ranges.
- State is derived from a city-to-state mapping (Appendix A).

4. Data Cleaning & Preparation

Table 4. Data-cleaning steps

Issue	Action taken
Non-UTF-8 characters in names	File read with Latin-1 encoding
9 missing cuisine values	Labelled "Not Specified" and excluded from cuisine counts
Duplicates	None found (0 duplicate rows or IDs); chain outlets kept
15 countries, 12 currencies	Kept India only: 8,652 rows in rupees
2,139 ratings of 0 ("Not rated")	Treated as unrated; rating statistics use rated restaurants only
9 restaurants with cost Rs.0	Treated as missing for price statistics
Extra spaces; Yes/No text	Trimmed text; converted to True/False
No geography or band fields	Added State, Region (NCR / rest of India), Cost Band, Rating Band
Outliers (cost up to Rs.8,000; votes up to 10,934)	Kept as genuine; medians and log scales used where needed

5. Descriptive Analysis

5.1 Overall aggregates

Table 5. Aggregate summary (India)

Measure	Value	Measure	Value
Restaurants	8,652	Total votes	1,187,163
States / cities	22 / 43	Rated restaurants	75.3%
Distinct cuisines	90	Online delivery	28.0%
Mean rating (rated)	3.35	Table booking	12.8%
Median cost for two	Rs.450	Median votes	24

5.2 State-wise distribution

Listings are heavily concentrated: Delhi, Haryana and Uttar Pradesh (the NCR states) hold 93.0% of all restaurants (Figure 1, Table 6). Every other state has fewer than 100 listings.

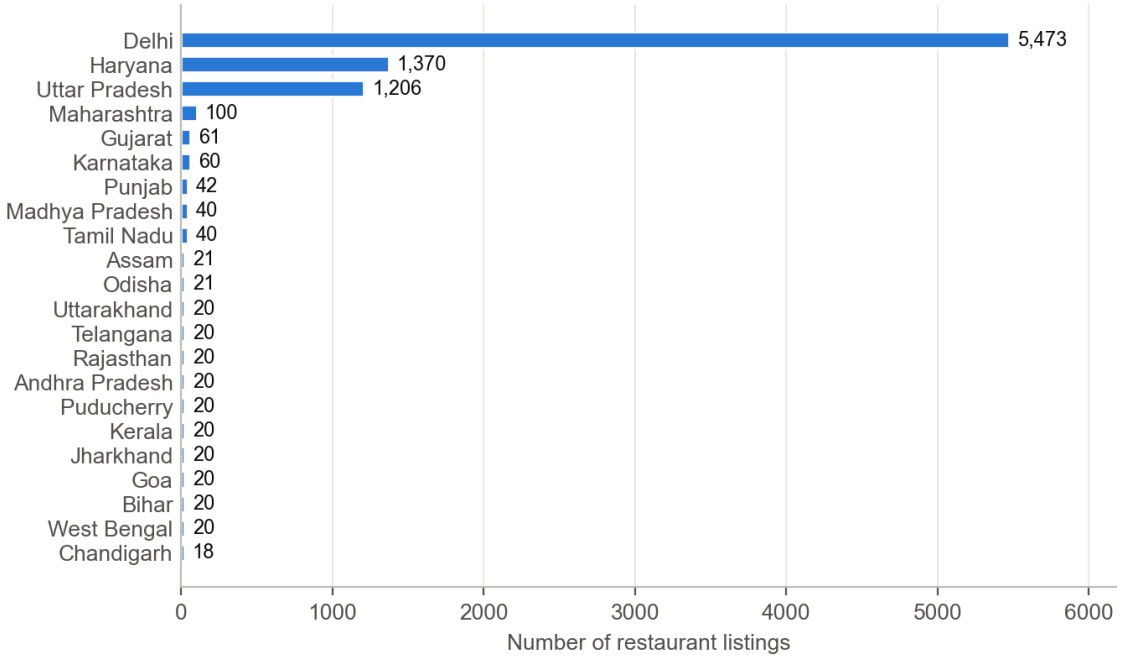


Figure 1. Number of restaurants by state / UT.

Table 6. State-wise aggregation (top 10 states)

State / UT	Listings	Share	Mean rating	Median cost (Rs.)	Online delivery
Delhi	5,473	63.3%	3.30	400	27%
Haryana	1,370	15.8%	3.30	500	34%
Uttar Pradesh	1,206	13.9%	3.23	500	31%
Maharashtra	100	1.2%	3.83	825	24%
Gujarat	61	0.7%	4.04	800	18%
Karnataka	60	0.7%	3.94	800	12%
Punjab	42	0.5%	3.84	750	2%
Madhya Pradesh	40	0.5%	3.96	650	0%
Tamil Nadu	40	0.5%	4.22	850	50%
Assam	21	0.2%	4.19	700	0%
Other 12 states/UTs	239	2.8%	–	–	–
India	**8,652**	100%	3.35	450	28%

States outside NCR show higher mean ratings and prices (Figure 2) because their small samples are mostly popular restaurants. Online delivery is uneven: 9 states have no delivery restaurant in the data. At city level, New Delhi (5,473) has almost five times as many listings as Gurgaon (1,118) (Figure 3).

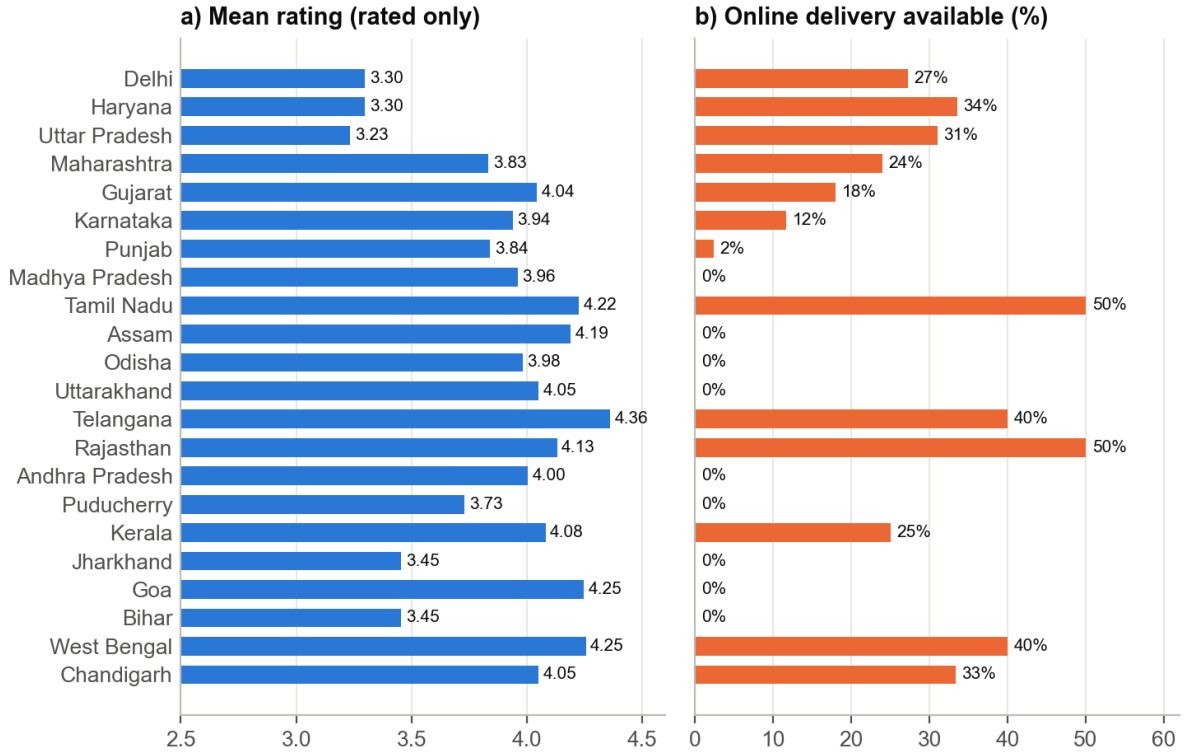


Figure 2. Mean rating (a) and share of restaurants with online delivery (b), by state.

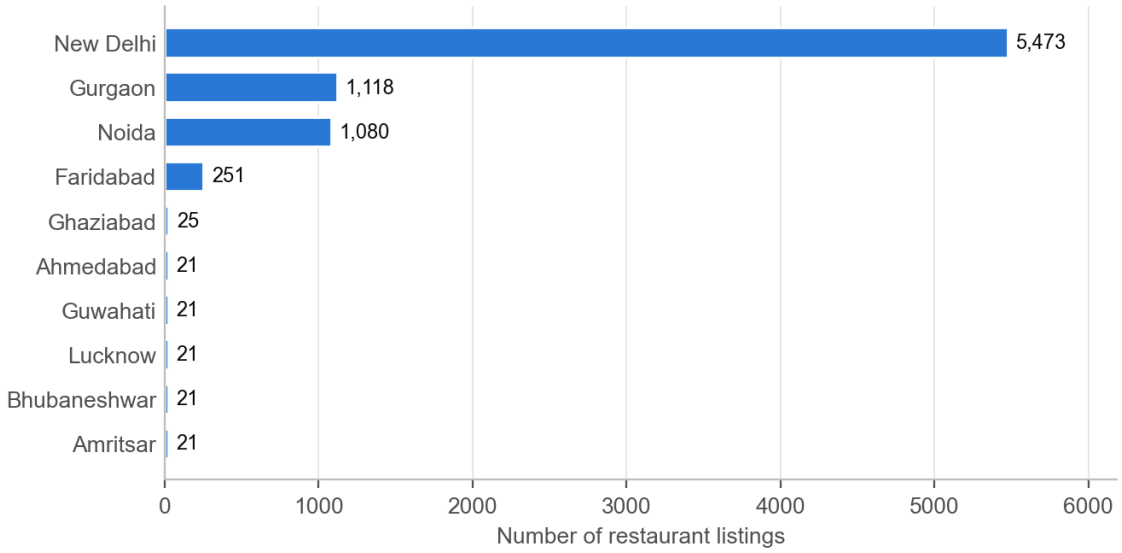


Figure 3. Top 10 cities by number of listings.

5.3 Cuisine

The 8,652 restaurants list 17,854 cuisine tags across 90 cuisines (median 2 per restaurant). North Indian is the most common cuisine, followed by Chinese and Fast Food (Figure 4).

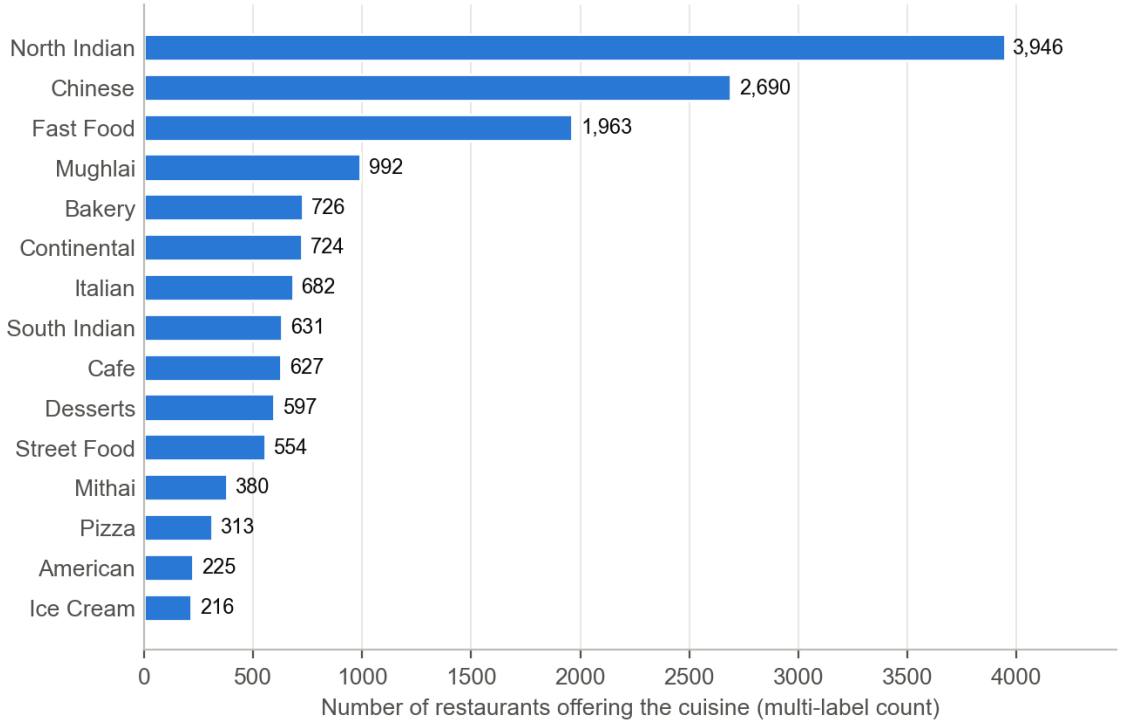


Figure 4. The 15 most common cuisines (a restaurant can list several).

Table 7. Frequency distribution and profile of the top 10 cuisines

Cuisine	Listings	% of restaurants	Median cost (Rs.)	Mean rating	Online delivery
North Indian	3,946	45.6%	500	3.29	29%
Chinese	2,690	31.1%	550	3.27	33%
Fast Food	1,963	22.7%	400	3.25	34%
Mughlai	992	11.5%	600	3.27	40%
Bakery	726	8.4%	350	3.37	26%
Continental	724	8.4%	1,100	3.70	30%
Italian	682	7.9%	1,000	3.71	41%
South Indian	631	7.3%	400	3.23	22%
Cafe	627	7.2%	600	3.63	30%
Desserts	597	6.9%	400	3.50	33%

Continental and Italian restaurants cost about twice the overall median and have the highest ratings (about 3.7). Street Food and Mithai are the cheapest (median Rs.200) (Figure 5).



Figure 5. Median cost for two (a) and mean rating (b) for the 12 most common cuisines.

5.4 Rating

Table 8. Measures of central tendency

Variable	n	Mean	Median	Mode
Rating (rated only)	6,513	3.35	3.3	3.1
Rating (incl. 0 = unrated)	8,652	2.52	3.1	0.0
Cost for two (Rs.)	8,643	624	450	500
Votes	8,652	137.2	24	0

Table 9. Measures of spread

Variable	Min	Max	Range	IQR	Variance	Std. dev.	Skew
Rating (rated)	1.80	4.90	3.10	0.70	0.25	0.50	0.23
Cost for two (Rs.)	50	8,000	7,950	400	354,794	596	3.59
Votes	0	10,934	10,934	96	183,218	428	9.66

Ratings are nearly symmetric: the mean (3.35), median (3.3) and mode (3.1) are close together, and the middle half of restaurants are rated 3.0–3.7 (Figure 6). Including unrated zeros would pull the mean down to 2.52. "Average" is the most common label (42.5%), and only 1.3% of restaurants are "Excellent" (Figure 7).

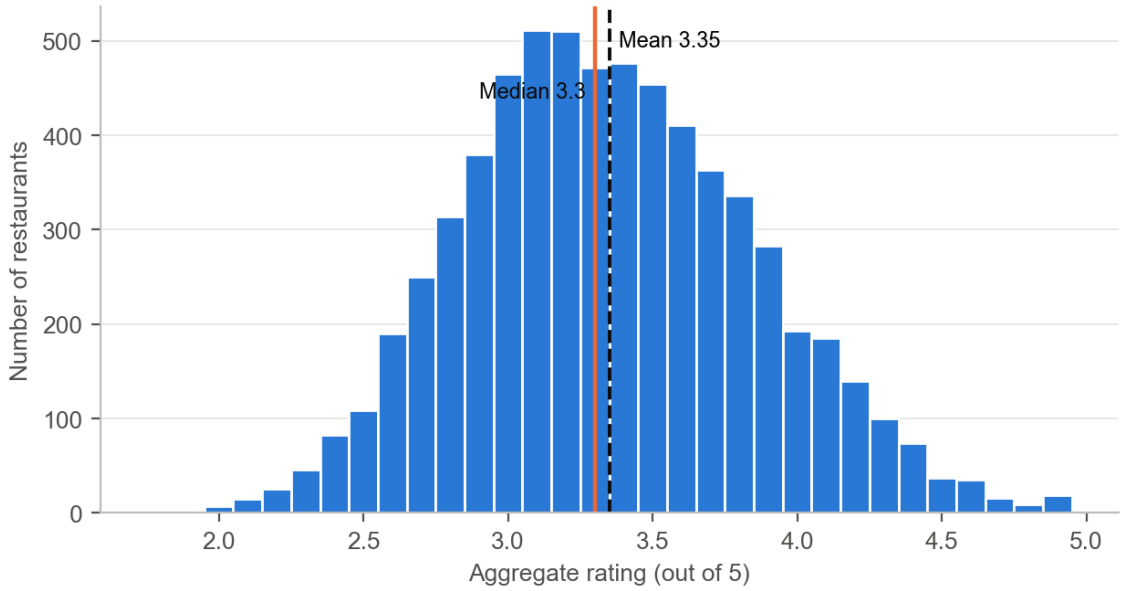


Figure 6. Distribution of ratings for rated restaurants (dashed line = mean, solid line = median).

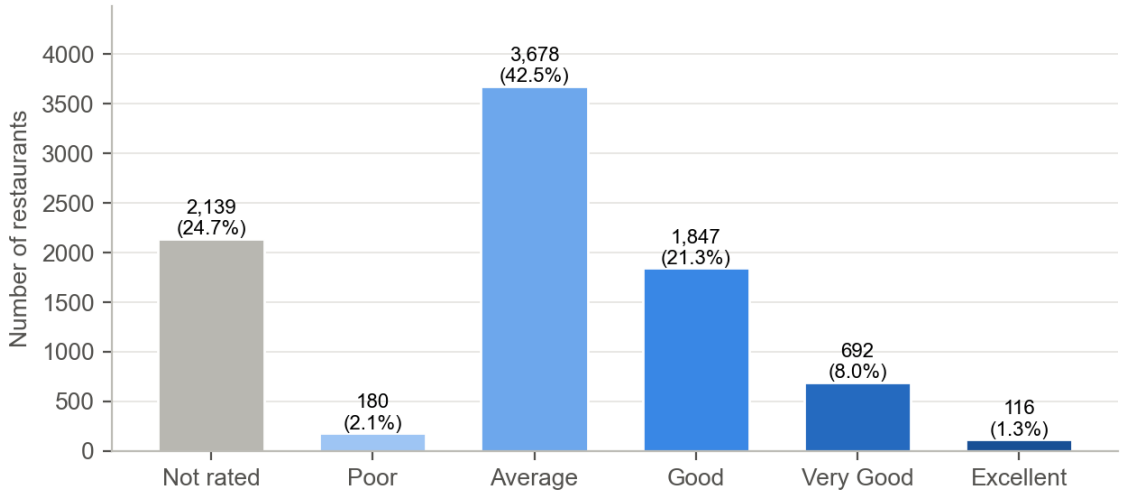


Figure 7. Frequency of Zomato rating categories.

5.5 Pricing

Cost for two is strongly right-skewed (skew 3.59): the mean (Rs.624) is above the median (Rs.450) because of a small premium tail (Figure 8). 81.7% of restaurants charge Rs.800 or less (Table 10).

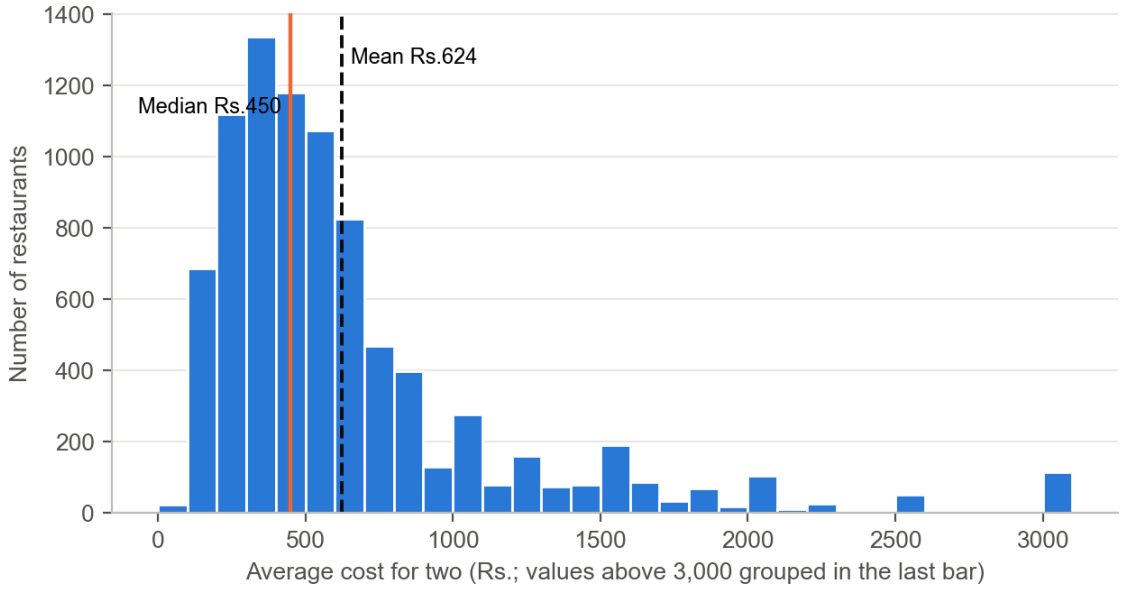


Figure 8. Distribution of average cost for two (values above Rs.3,000 grouped in the last bar).

Table 10. Frequency distribution by cost band

Cost band (Rs.)	Listings	Relative freq.	Cumulative	Mean rating	Online delivery
<=300	2,716	31.4%	31.4%	3.22	11%
301-500	2,523	29.2%	60.6%	3.22	30%
501-800	1,823	21.1%	81.7%	3.32	46%
801-1200	684	7.9%	89.6%	3.57	43%
1201-2000	674	7.8%	97.4%	3.77	34%
>2000	223	2.6%	100.0%	3.68	8%

Zomato's price ranges are well separated: the median cost roughly doubles at each step, from Rs.300 to Rs.2,200 (Figure 9). Half of all listings are budget restaurants (Figure 10).

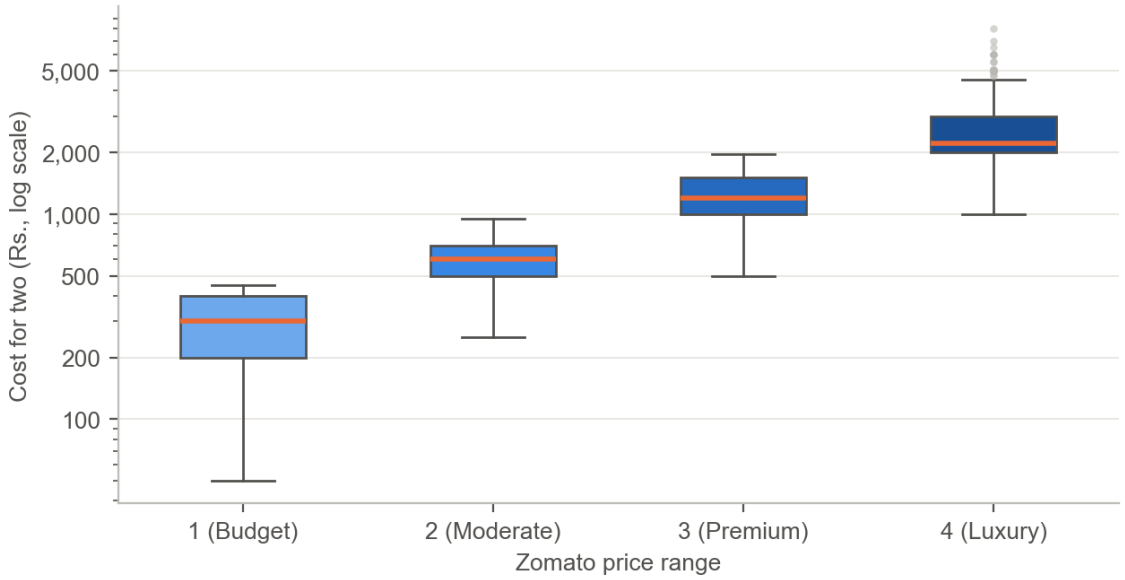


Figure 9. Cost for two within each price range (log scale; orange line = median).

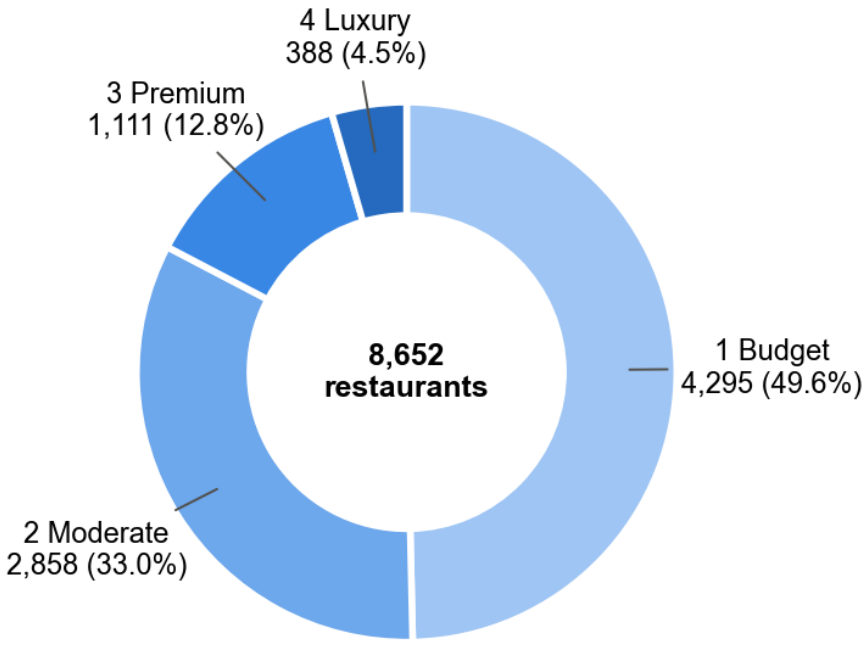


Figure 10. Share of restaurants in each price range.

5.6 Online delivery

2,423 restaurants (28.0%) offer online delivery. It is most common in the moderate range and rare in the budget and luxury ranges (Figure 11).

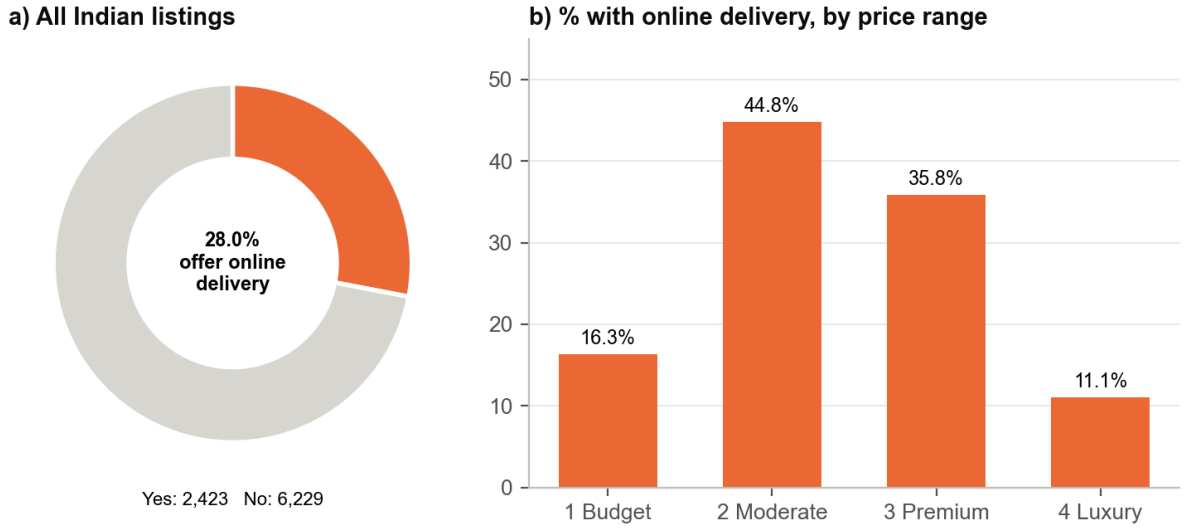


Figure 11. Online-delivery share overall (a) and by price range (b).

Table 11. Restaurants with vs. without online delivery

Measure	With delivery	Without delivery
Restaurants	2,423	6,229
Rated (% of group)	96.0%	67.2%
Mean rating (rated)	3.37	3.34
Median cost for two (Rs.)	600	400
Median votes	77	13

Delivery restaurants are almost all rated (96.0%) and have about 6 times the median votes, but their mean rating is about the same (3.37 vs 3.34). Delivery goes with **more** feedback, not **better** feedback (Figure 12).

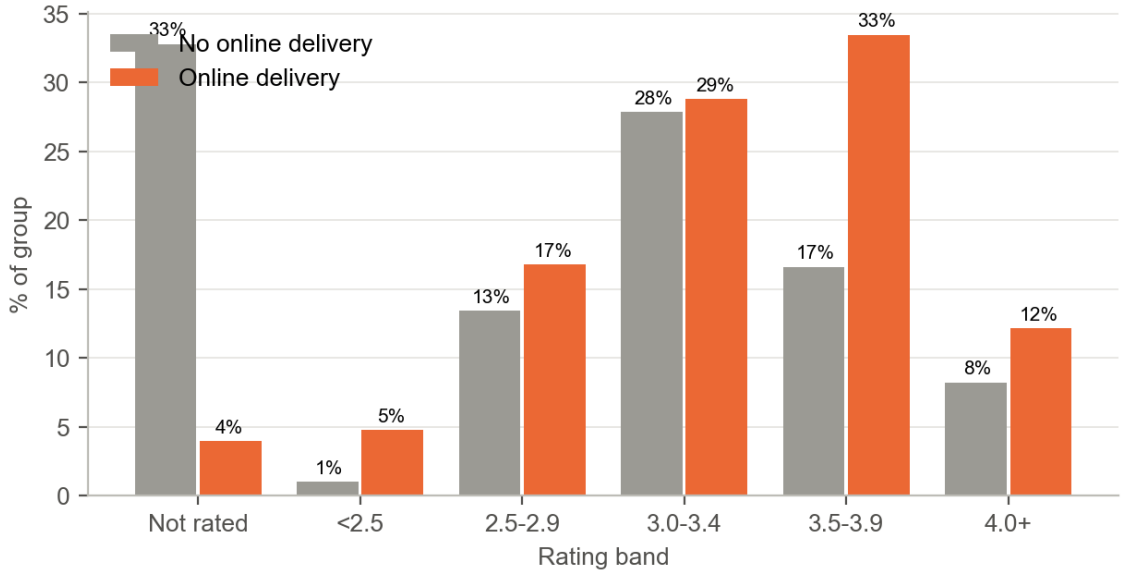


Figure 12. Rating-band distribution for restaurants with and without online delivery.

5.7 Cross-tabulation

Table 12. Cross-tabulation: price range × rating band (counts)

Price range	Not rated	<2.5	2.5-2.9	3.0-3.4	3.5-3.9	4.0+	Total
1 – Budget	1,699	61	652	1,232	552	99	4,295
2 – Moderate	396	96	494	910	709	253	2,858
3 – Premium	33	18	84	218	419	339	1,111
4 – Luxury	11	5	13	75	167	117	388
Total	2,139	180	1,243	2,435	1,847	808	**8,652**

39.6% of budget restaurants are unrated, compared with 3.0% of premium ones, while 30.5% of premium restaurants are rated 4.0 or above (Figure 13).

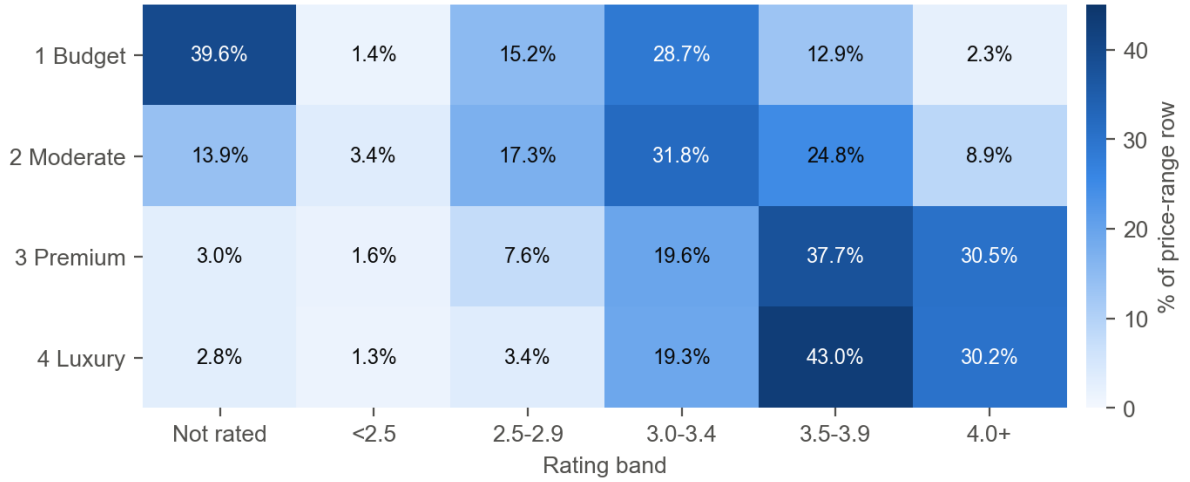


Figure 13. Heat map of price range × rating band (row %).

Table 13. Cross-tabulation: online delivery × table booking

	Booking: Yes	Booking: No	Total
Delivery: Yes	427	1,996	2,423
Delivery: No	684	5,545	6,229
Total	1,111	7,541	8,652

5.8 Trend (progression) and engagement

The data has no dates, so the analysis tracks metrics across the ordered price ranges instead. From budget to luxury, the mean rating rises from 3.20 to 3.73, table booking from 0% to 65%, and median votes from 7 to 180 at the premium range (Figure 14). Engagement also rises with rating (Figure 15) and is highly concentrated: the top 20% of restaurants receive 83.6% of all votes (Figure 17).

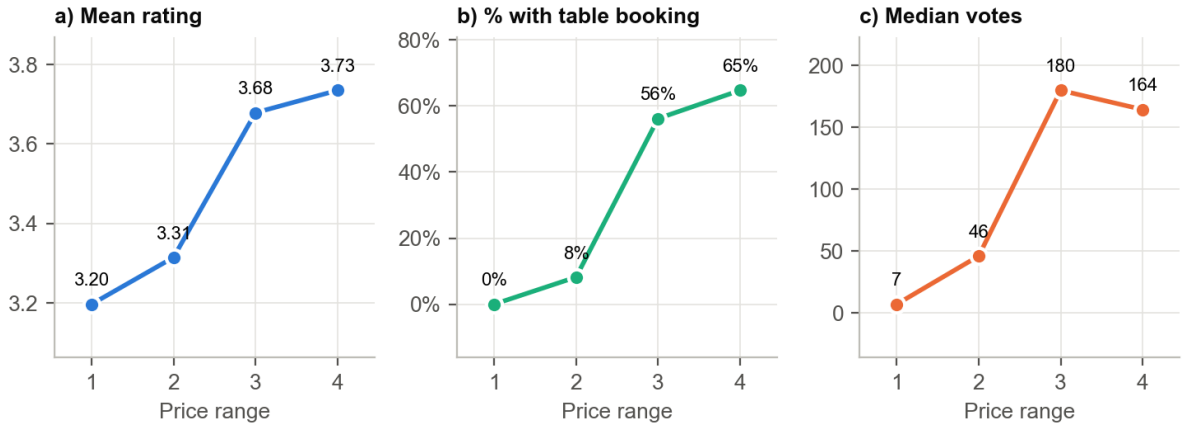


Figure 14. Mean rating (a), table-booking share (b) and median votes (c) across price ranges.

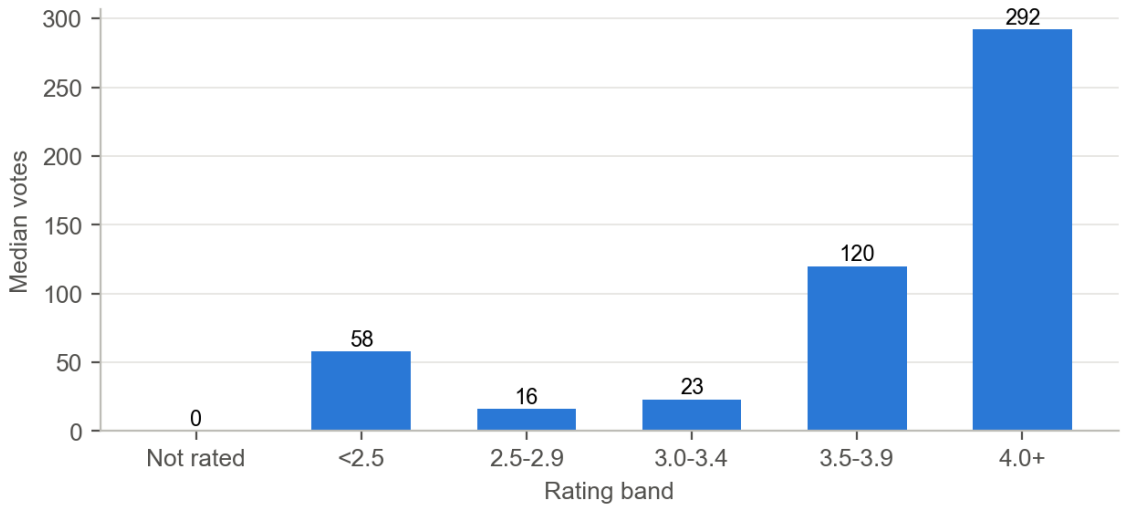


Figure 15. Median votes in each rating band.

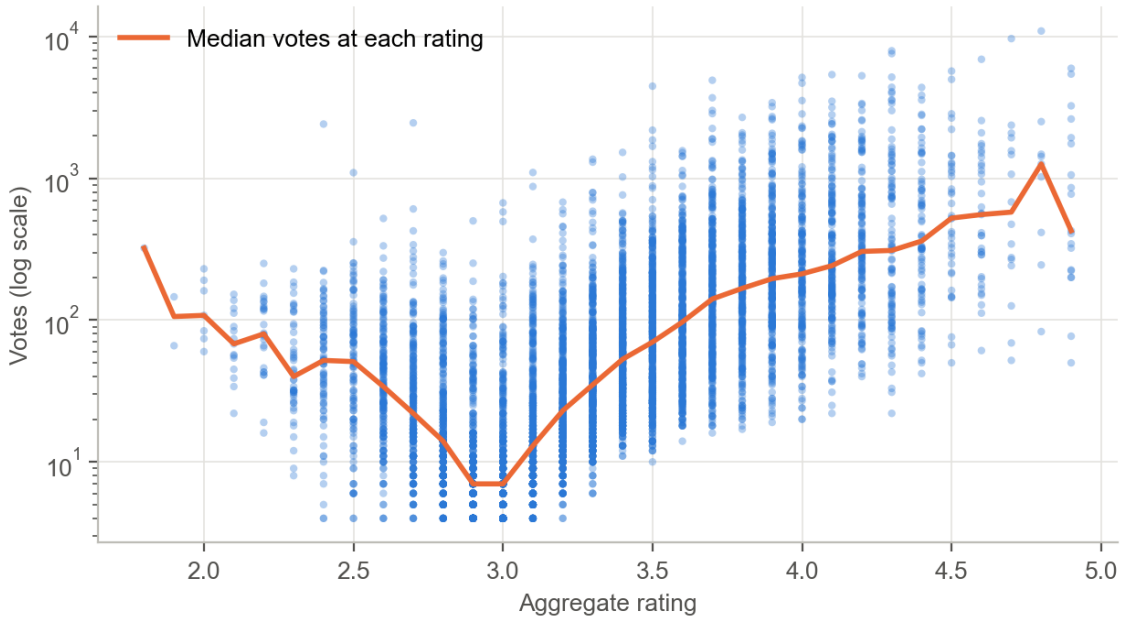


Figure 16. Votes vs. rating for rated restaurants (log scale), with the median votes at each rating.

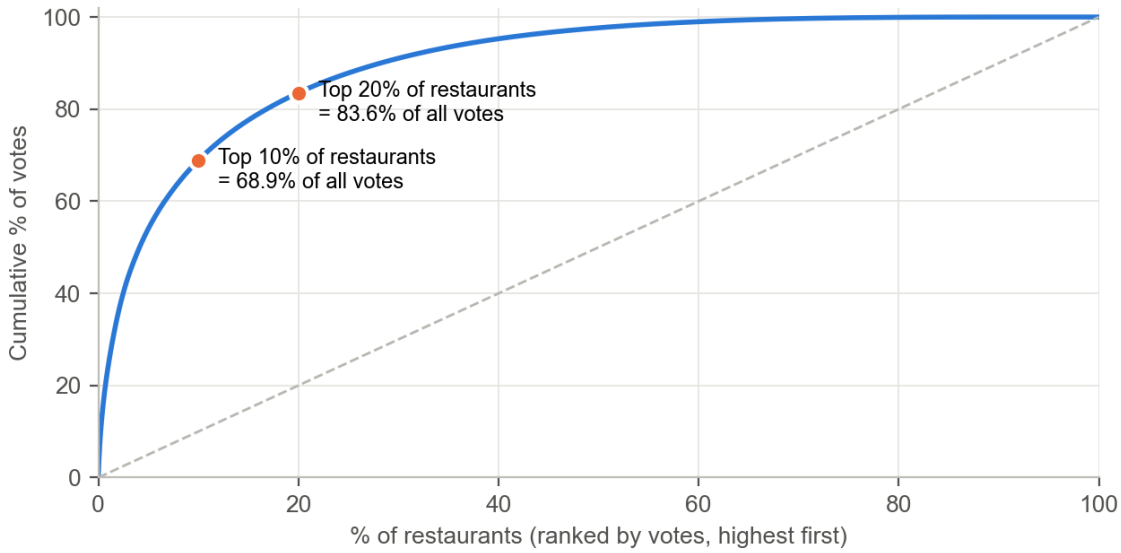


Figure 17. Cumulative share of votes by restaurants ranked from most to least voted.

6. Data Visualization: Summary Dashboard

Figure 18 combines the main measures into one dashboard. Section 5 also contains 17 charts, each placed next to the statistics it illustrates: bar charts for categories, histograms for distributions, a box plot for spread, donuts for shares, line charts for progression and a heat map for the cross-tabulation.

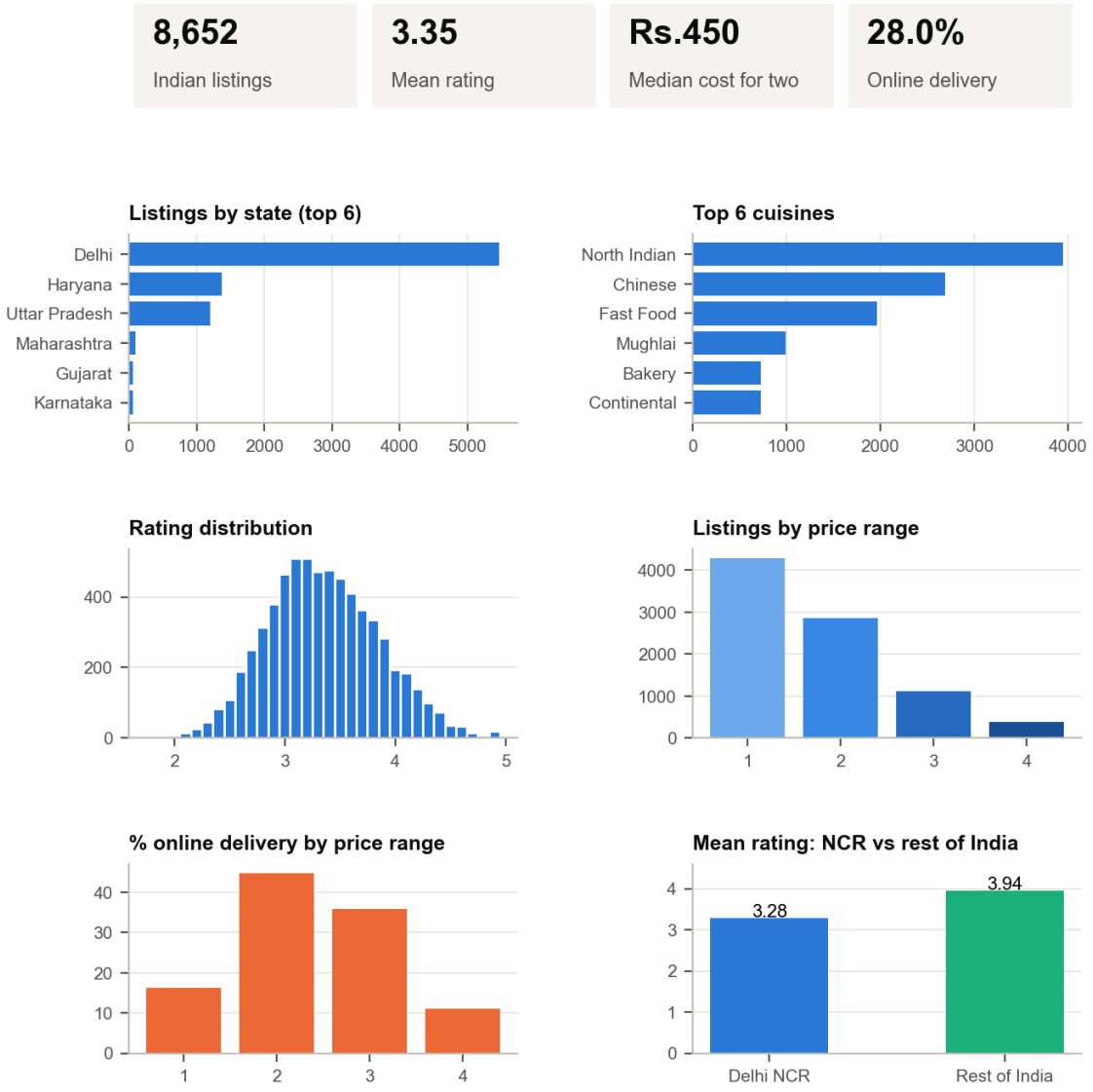


Figure 18. Summary dashboard: KPIs and the state, cuisine, rating, price-range, delivery and regional views.

7. Key Findings

Table 14. Summary of key findings

Theme	What the data shows
State-wise	91.9% of listings are in Delhi NCR; other states are thinly covered and 9 have no delivery restaurants.
Cuisine	North Indian, Chinese and Fast Food dominate; Continental and Italian are the premium, highest-rated niche.
Rating	Symmetric around 3.3 (SD 0.50); "Average" is the modal label; 24.7% are unrated.
Pricing	Affordable (median Rs.450) but right-skewed; higher price goes with higher rating.
Online delivery	28.0% of restaurants; peaks in the moderate range; linked to far more votes but not higher ratings.
Engagement	Top 10% of restaurants hold 68.9% of votes; votes rise with rating and price.

8. Recommendations

- **Extend online delivery to budget restaurants:** only 16.3% offer it, although they make up half of all listings.
- **Reduce unrated listings (24.7%):** prompt customers for first reviews.
- **Broaden coverage outside Delhi NCR**, especially delivery in states with none.
- **Feature high-rated cuisines** (Continental, Italian, Cafe) in curated collections.
- **Report medians, not means,** for price and votes, because both are skewed.
- **Collect order-level data** (orders, value, time) to allow true sales and trend analysis.

9. Conclusion

Using 8,652 real Zomato listings, this report applied aggregation, central tendency, spread, frequency distributions, cross-tabulation and progression analysis, with 18 charts. The data shows a market concentrated in Delhi NCR and built on North Indian, Chinese and fast food. Ratings are moderate and prices are mostly affordable but skewed. Online delivery reaches just over a quarter of restaurants and is closely linked with customer engagement. These findings describe what the data shows; explaining or forecasting them would need predictive analysis and order-level sales data.

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Appendix A: City-to-State Mapping

Table A1. Indian cities, mapped state and number of listings

City	State / UT	n	City	State / UT	n
New Delhi	Delhi	5,473	Mangalore	Karnataka	20
Gurgaon	Haryana	1,118	Agra	Uttar Pradesh	20
Noida	Uttar Pradesh	1,080	Kochi	Kerala	20
Faridabad	Haryana	251	Kolkata	West Bengal	20
Ghaziabad	Uttar Pradesh	25	Dehradun	Uttarakhand	20
Ahmedabad	Gujarat	21	Allahabad	Uttar Pradesh	20
Guwahati	Assam	21	Aurangabad	Maharashtra	20
Lucknow	Uttar Pradesh	21	Bangalore	Karnataka	20
Bhubaneshwar	Odisha	21	Bhopal	Madhya Pradesh	20
Amritsar	Punjab	21	Chennai	Tamil Nadu	20
Pune	Maharashtra	20	Coimbatore	Tamil Nadu	20
Puducherry	Puducherry	20	Goa	Goa	20
Patna	Bihar	20	Indore	Madhya Pradesh	20
Ludhiana	Punjab	20	Jaipur	Rajasthan	20
Ranchi	Jharkhand	20	Kanpur	Uttar Pradesh	20
Surat	Gujarat	20	Vizag	Andhra Pradesh	20
Vadodara	Gujarat	20	Chandigarh	Chandigarh	18
Nashik	Maharashtra	20	Hyderabad	Telangana	18
Nagpur	Maharashtra	20	Secunderabad	Telangana	2
Mysore	Karnataka	20	Panchkula	Haryana	1
Mumbai	Maharashtra	20	Mohali	Punjab	1
Varanasi	Uttar Pradesh	20			